GLOSSIFUNGITES ICNOFACIES RELATED TO STRATIGRAPHIC DISCONTINUITIES OF MESOZOIC–CENOZOIC DEPOSITS IN THE ARAUCO BASIN, CENTRAL CHILE

Patricio Zambrano-Lobos[1]*; Martin Jara Cortes[1]; Carlos Asenjo Fernandoy[1]; Alfonso Encinas[2]

[1]Septos Asesorías Geológicas, Chile. **patricio.zambrano@septos.com*
[2]Universidad de Concepción, Chile. *aencinas@udec.cl*

The Arauco Basin (Central Chile) is located in the coastal forearc domain of the Andean active margin of the South American Plate and corresponds to a part of the raised platform. It hosts a ~2300 m thick Maastrichtian–Holocene sedimentary succession [[1]]. Although sedimentological characterizations of stratigraphic units have been relatively well studied, the contact relationships between these units lack research.

Within the Arauco Basin, five discordant tectonostratigraphic sequences have been recognized [[1]]. From base to top, these are: (S1) Late Cretaceous (Santonian-Maastrichtian); (S2) Paleocene-Middle Eocene; (S3) Lower Miocene; (S4) Pliocene-Pleistocene; and (S5) Pleistocene-Holocene, which correlate to the Quiriquina Formation, Lebu Group, Ranquil Formation, Tubul Formation, and Cañete Formation, respectively. Each of these sequences is separated by sedimentary hiatuses spanning different time intervals [[3]]. In this context, this study focuses on the trace fossils recognized from the four discontinuities throughout the Mesozoic–Cenozoic. The base of the entire sedimentary sequence overlies an unconformity to Paleozoic metamorphic basement rocks, in which *Gastrochaenolites* isp. have been reported [[2]].

Four discontinuity surfaces were recognized, characterized by the following features: SB1, local colonization with *Gastrochaenolites ornatus* and *Rhizocorallium* isp.; SB2, *Gastrochaenolites*; SB3, scarce and local presence of *Gastrochaenolites*; and SB4, local presence of *Gastrochaenolites* in bimodal sizes. Generally, the specimens recognized on the discontinuity surfaces exhibit passive infill and vertical disposition. The lithologies associated with the discontinuities range from shale to medium-grained sandstone. All the colonized surfaces here studied, was generated in a coastal marine environment during an important sea-level fall leading to the exhumation and erosion, conforming a sequence boundary. Subsequently, during a sea level rise, these surfaces, constituted as firmgrounds, were colonized during the early periods of transgression, giving rise to the conditions required for the development of the *Glossifungites* Ichnofacies. Differences in size and diameter of *Gastrochaenolithes* suggest a complex colonization history associated with paleoenvironmental stress.

REFERENCES

[[1]] Becerra, J.; Contreras-Reyes, E.; Arriagada, C. 2013. Seismic structure and tectonics of the southern Arauco Basin, south-central Chile (~ 38 S). *Tectonophysics*, v. 592, p. 53–66.
[[2]] Buatois, L.A.; Encinas, A. 2011. Ichnology, sequence stratigraphy and depositional evolution of an Upper Cretaceous rocky shoreline in central Chile: Bioerosion structures in a transgressed metamorphic basement. *Cretaceous Research*, v. 32, p. 203–212.
[[3]] Kuhn, P.P.; Echtler, H.P.; Littke, R.; Alfaro, G. 2010. Thermal basin modelling of the Arauco forearc basin, south central Chile—heat flow and active margin tectonics. *Tectonophysics,* v. 495, p. 111–128.